USO
just crunch it

Cursul 10 bis



10 bis

Sisteme Integrate

4 decembrie 2017

Moto

We choose to go to the moon

John F. Kennedy

In saptamana aceasta la USO

- Laboratorul 10 – Shell Scripting incepe luni 4 decembrie
- Cursul 11 – Securitate va fi luni 11 decembrie
- Lucrarea 3 de curs va fi luni 11 decembrie
- Tema 3 se va incheia luni 4 decembrie
- Tema 4 (suplimentara/bonus) va fi marti 5 decembrie

Cuprins

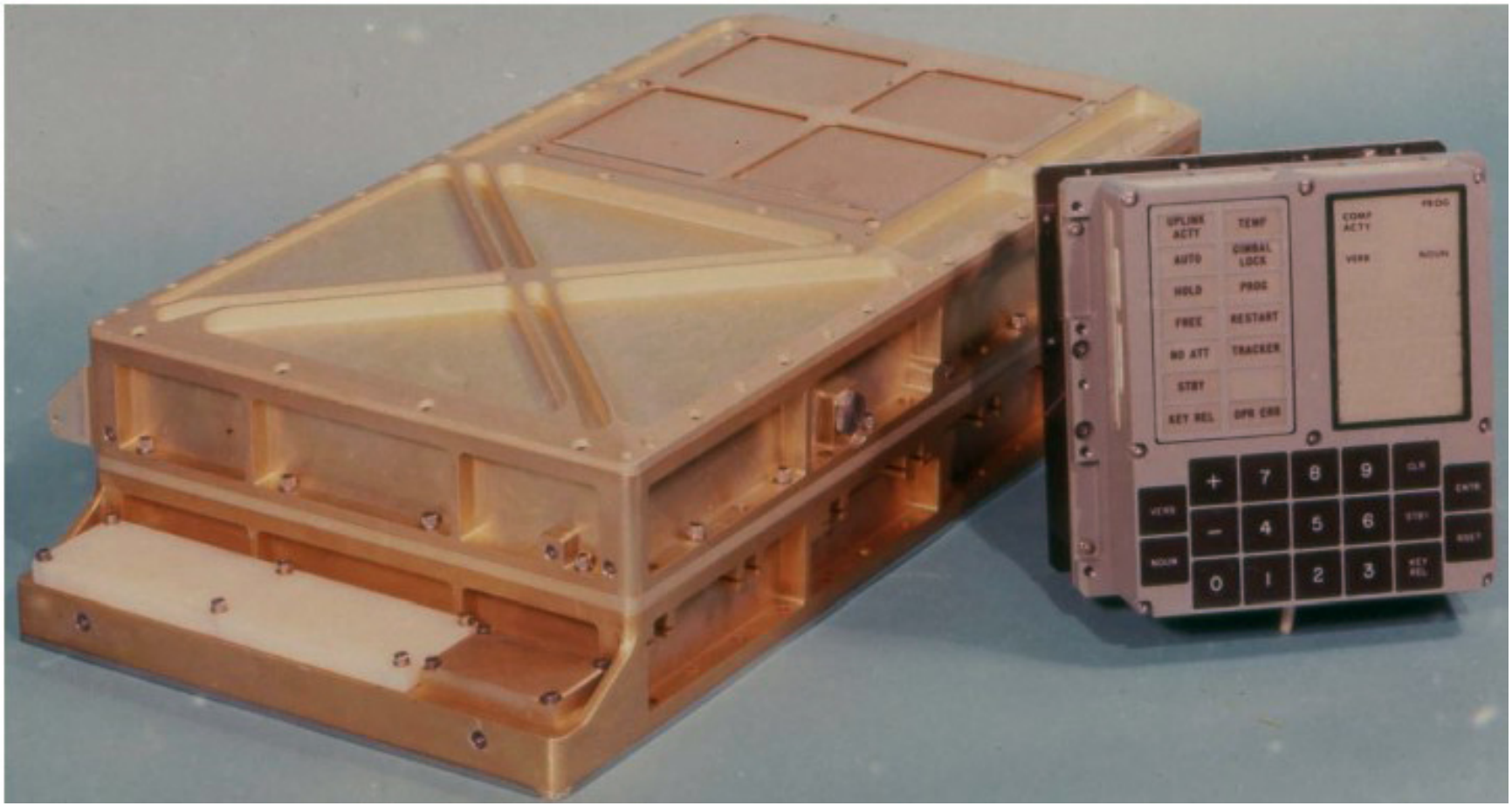
- Sisteme Integrate
- Exemple
- Propietati ale sistemelor integrate

- Calculatoare folosite pentru
  - Automatizari
  - Industrie
  - Control de sisteme
- Doua tipuri
  - Microcontroller
  - Calculator

Apollo Guidance Computer (DSKY)

- 2,048 MHz
- 4096 B RAM
- ~ 70 KB ROM
- Interfata - DSKY

Apollo Guidance Computer (AGC-DSKY)



Android

- Telefoane
- Ceasuri
- Masini
- Televizoare



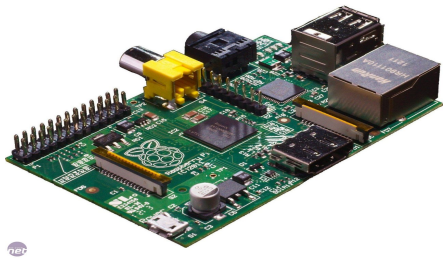
ANDROID

Microcontrollere si Calculatoare

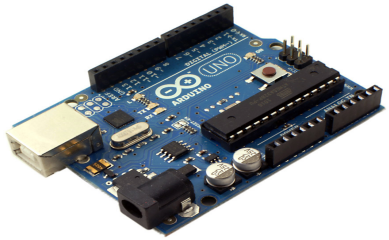
- Sisteme simple
- Control de hardware
- Viteza mica
- Memorie limitata
- GPIO
- PWM
- ADC
- Ruleaza un singur program,



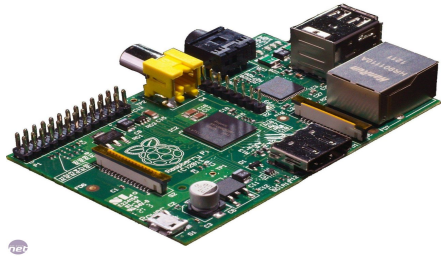
- Procesor normal
- Viteza mare
- Memorie multa
- GPIO
- Ruleaza un Sistem de operare



Microcontrollere si Calculatoare

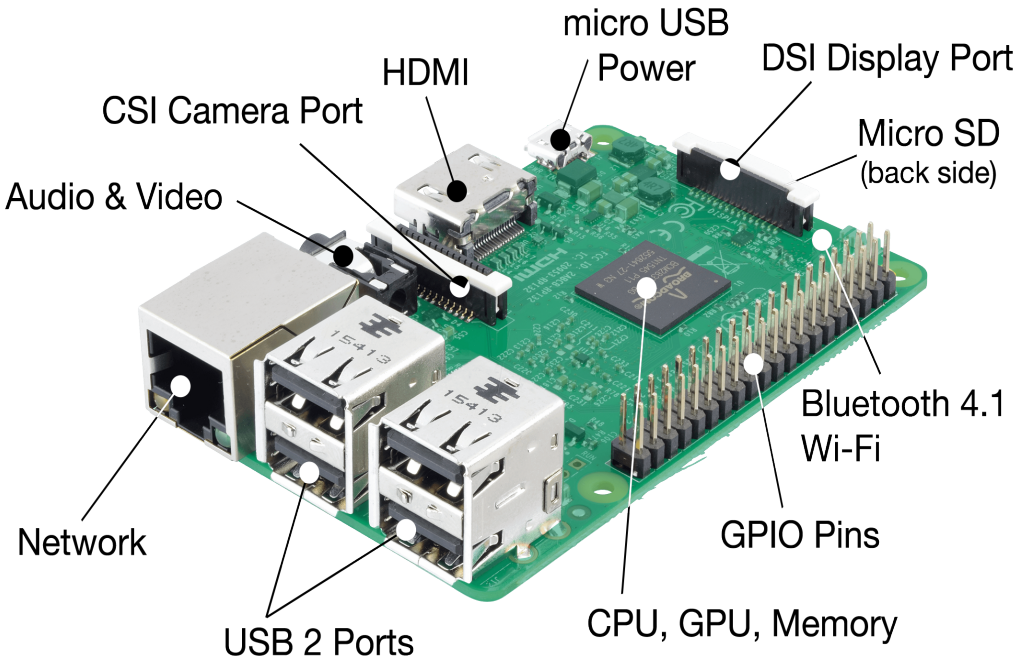


Firmware

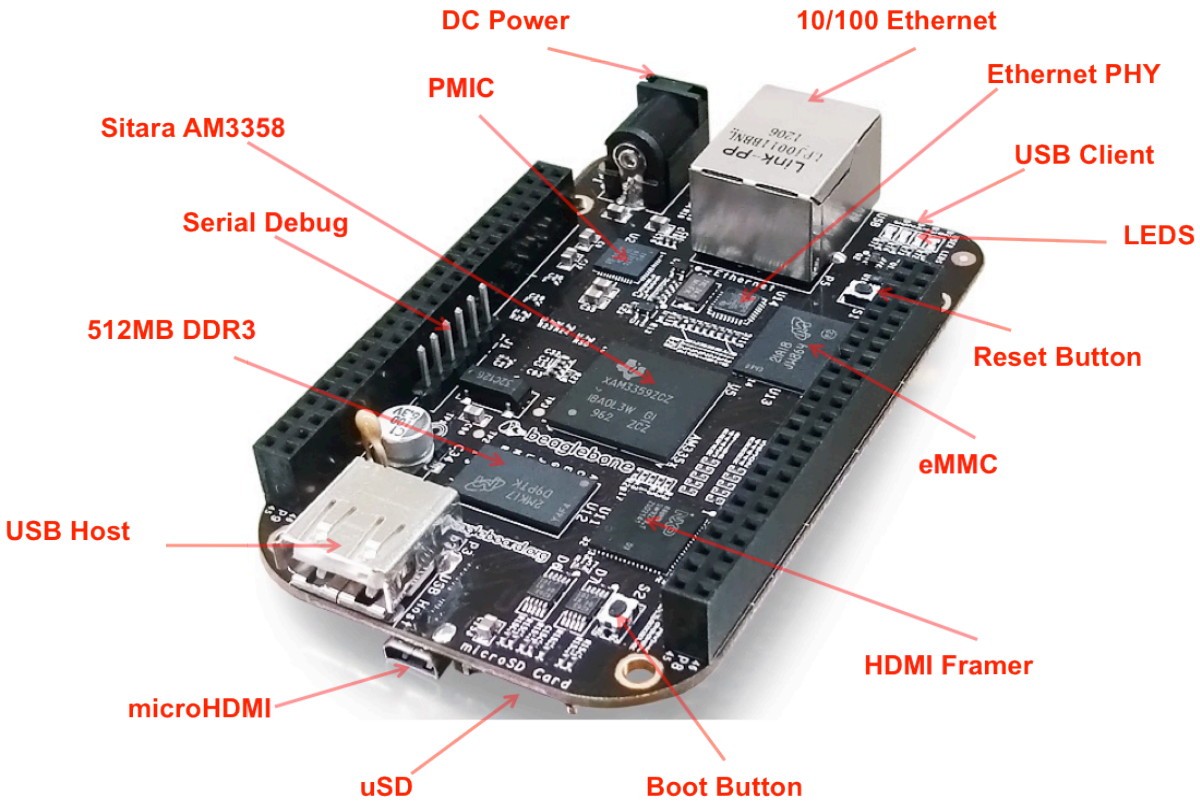


Software

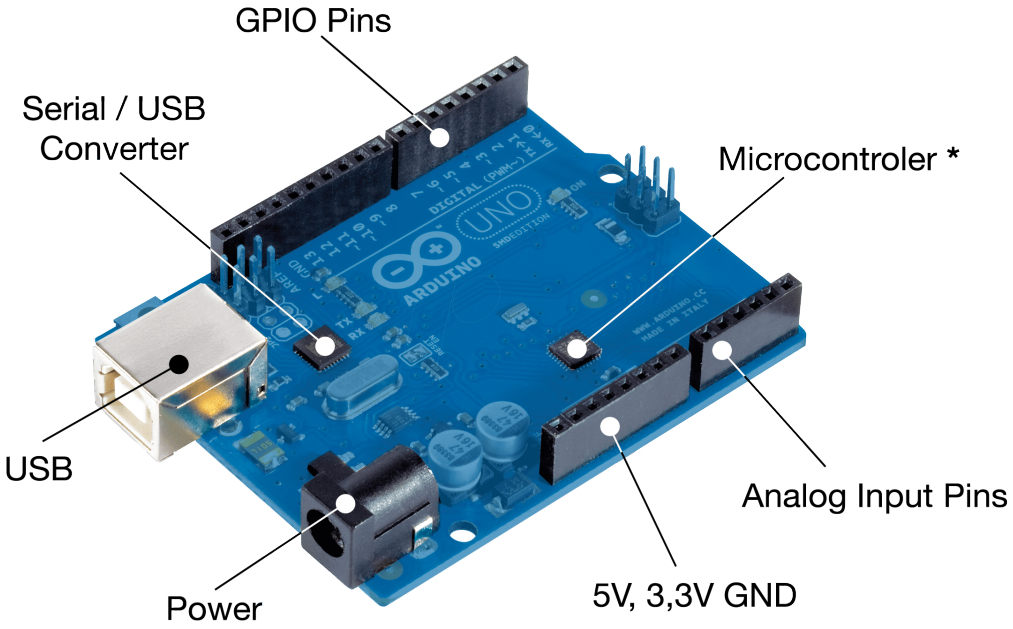
Raspberry Pi 3



BeagleBone Black



Arduino UNO Rev3 (SMD)



Elemente particulare pentru sisteme integrate

- Procesor (Socket on a chip - SoC)
- Boot
- Baterie
- Ecran
- Pini
- Update

Procesor

Calculatoare

- **ARM**
  - 32 biti - v6, v7
  - 64 biti - v8
- MIPS
  - In routere
- x86
  - Atom
  - Quark

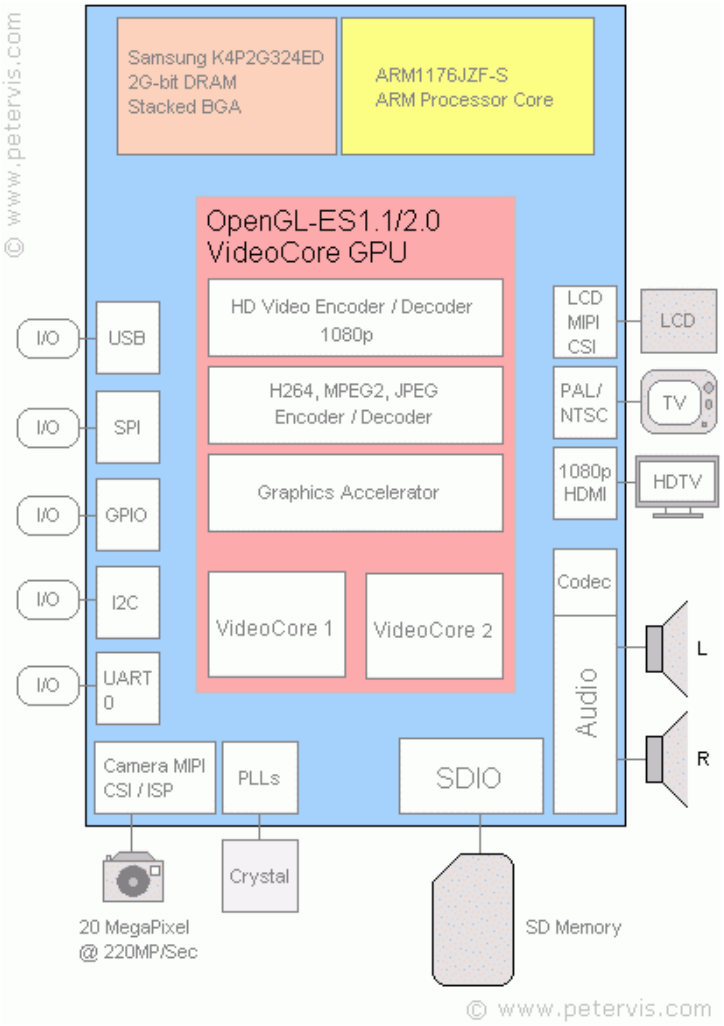
Microcontrollere

- **AVR**
  - ATmega
  - Attiny
- PIC
  - PIC8
  - PIC16
  - PIC32
- ARM
  - ARM Cortex M0, M3, M4

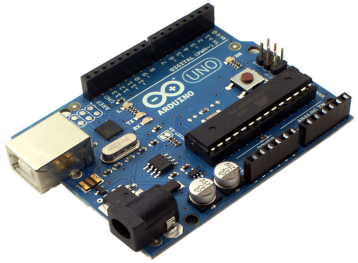
Socket on a chip - SoC

- Procesor
- Memorie
- Placa de baza
- Placa de sunet
- Placa video

Socket on a chip - SoC

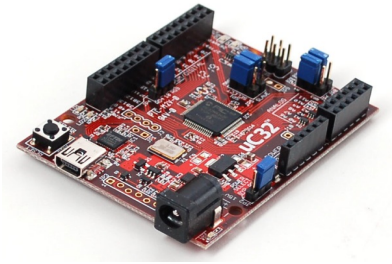


Microcontrollere



Arduino
$25
ATmega328

ChipKIT
$30
PIC





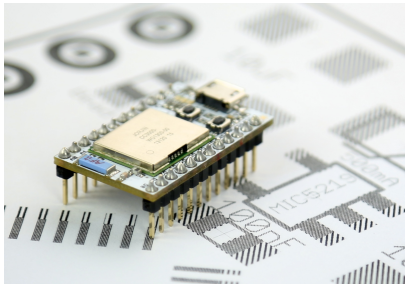
LaunchPad
$4
MSP430

Microcontrollere



STM32
$30
ARM Cortex M0,
M3, M4

Particle
$35
ARM
WiFi Internet





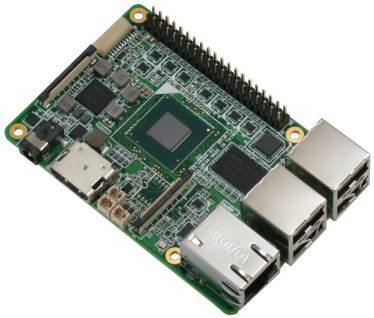
WiPy
$45
ESP32
WiFi Internet
Python

Procesare si retea



Raspberry Pi
$35
1000 MHz ARMv7, GPU
1 GB RAM
Compute Module

Up Board
$120
1.84 GHz Intel Atom
1, 2 sau 4 GB RAM

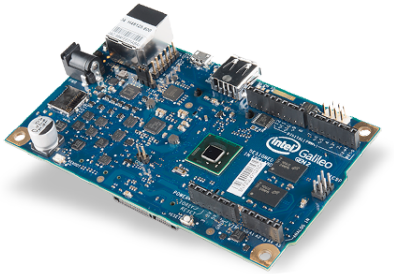




Beaglebone Black
$45
1 GHz ARM, GPU
512 MB RAM
4 GB Flash

Procesare si retea

Intel® Galileo
$50
400 MHz Quark x86
256 MB RAM

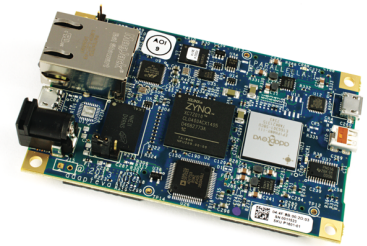




Intel® Edison
$70
1 GHz Dual Core Atom x86
1 GB RAM
WiFi
BLE
4 GB Flash

UDOO Neo
$50
i.MX 6 Solo ARM, GPU
ARM M4
512 MB or 1 GB RAM





Parallella
$99
1 GHz Dual Core Zynq ARM
16 or 64 Epiphany CPUs

Functii limitate

- Floating Point Unit
  - Soft Float - compilatorul simuleaza instructiunile folosind numere intregi
  - Hard Float – compilatoprul genereaza instructiuni pentru FPU
    - executate de sistemul de operare daca nu exista FPU pe sistem

Bootare

- Mai multe moduri de bootare
  - **Serial**
  - **SPI**
  - SD Card
  - USB
  - Retea
- Sistem de update integrat in boot
  - **Update automat**
  - Update manual
- Backup boot
- UBoot – sistem de bootare

Update

- Vor trebui update-uri
  - functii noi
  - probleme de securitate
- OS
  - doua partitii
- Applications
  - snap
  - Google Store (Android Things)



android things

Ecran

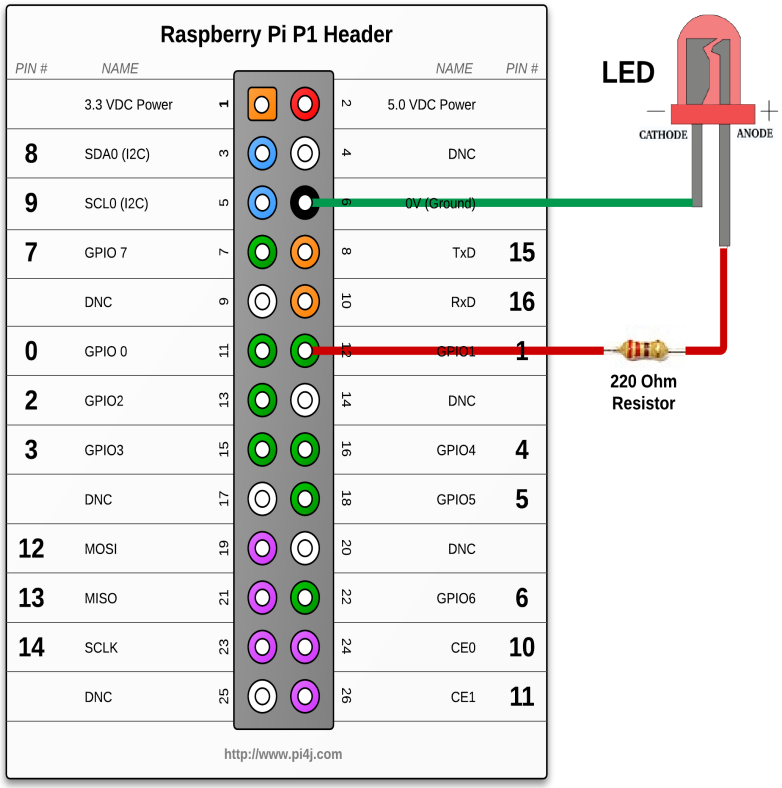
- Fara ecran
  - Router (interfata web)
- Ecran text
  - POS
- Ecran mic
  - Masina de spalat
- Ecran normal de dimensiuni mici
  - Cafetiere

Pini

- Pini Digitali
  - Valori LOW (0) sau HIGH (1)
- Pinii sunt
  - Baterii (OUTPUT)
  - Voltmetre (INPUT)



- Pinii se comporta ca o baterie
  - LOW – 0V
  - HIGH – 5V, 3.3V or 1.8V

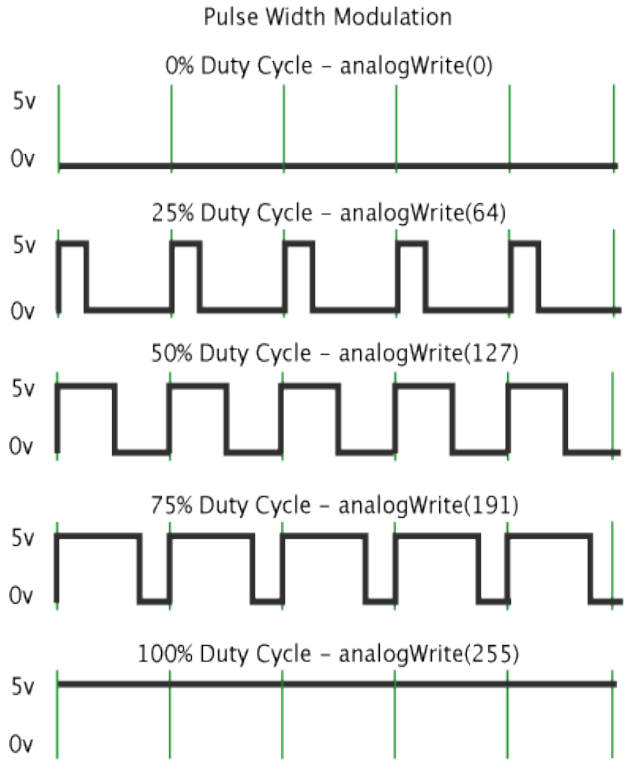


- Masuram tensiunea la borne



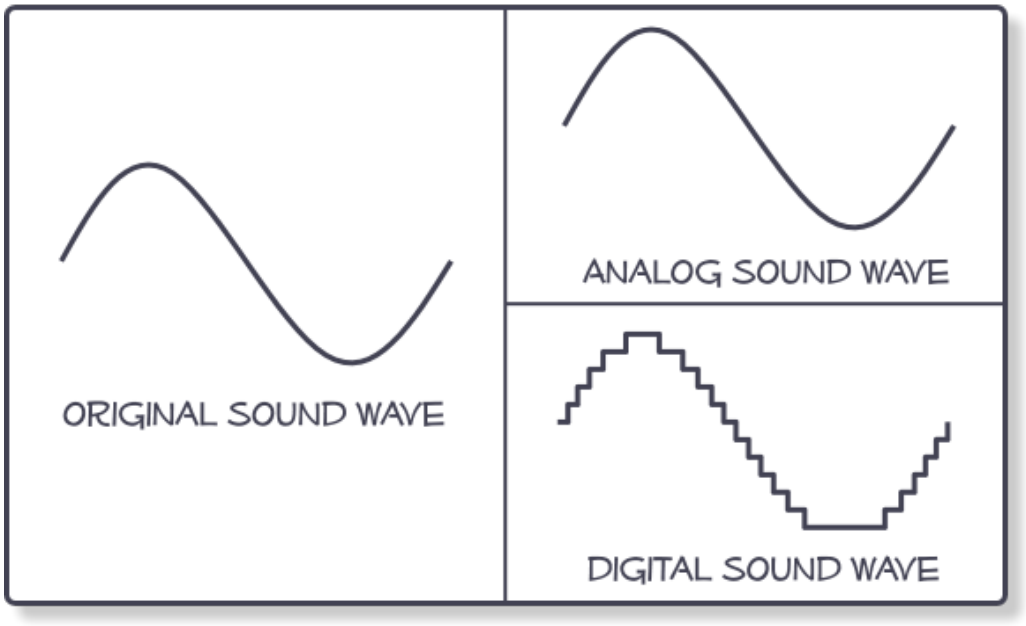
Modulare in latime de puls

- Setam % pentru “high” cycle
  - 0 – 0%
  - 255 – 100%
    - Depinde de biblioteca
- Implementare
  - Hardware
  - Software
- Utilizare
  - LED dimming
  - Servo Motors



Convertoare Analog Digitale

- Masoara tensiunea
- Valoare pe n biti
  - 0 - 1023



Eben Upton, CBE



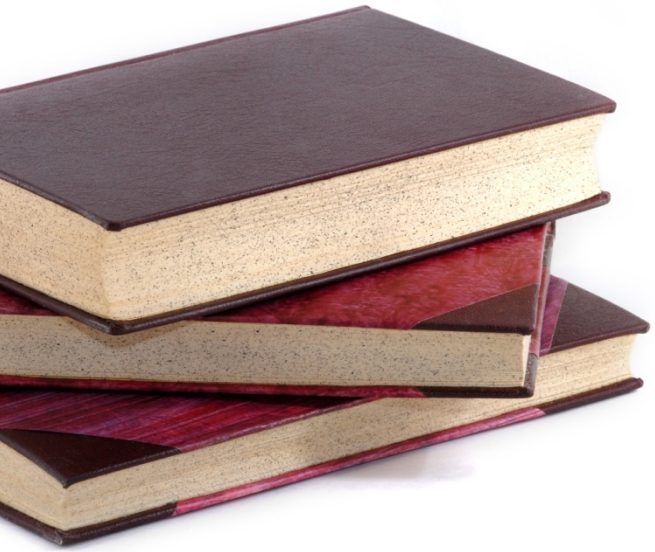
- Britanic
- University of Cambridge
- A lucrat la Boardcom
- Inventaorul Raspberry Pi

Distributii Linux pentru sisteme integrate

USB
just crunch it

android

Raspbian



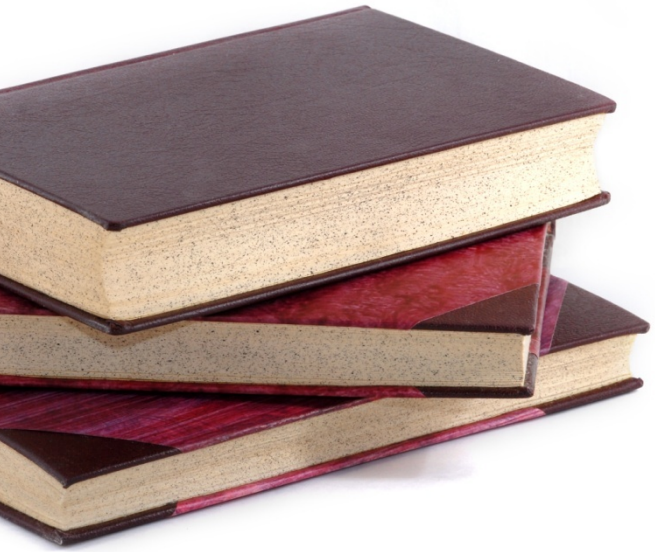
yocto
PROJECT

Linaro

android
things

Core

Producatori de sisteme integrate



just crunch it

ARDUINO

intel Quark

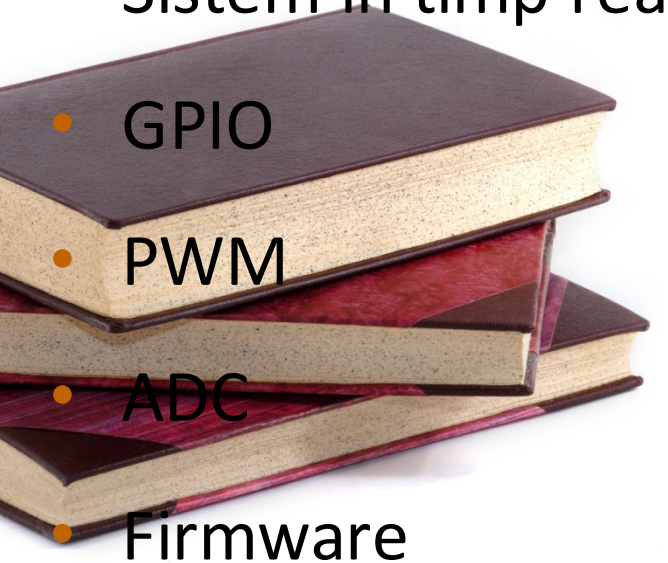
redpitaya

pandaboard.org

Qualcomm snapdragon

Cuvinte cheie

- Sisteme integrate
- Calculatoare
- Microcontrollere
- Sistem in timp real
- GPIO
- PWM
- ADC
- Firmware



- Software
- Update
- Uboot
- SoC
- ARM
- PIC
- AVR
- MIPS

Link-uri utile

- http://www.raspberrypi.org
- http://www.arduino.cc
- http://beagleboard.org/
- http://www.up-board.org/
- http://w3.siemens.com/mcms/sce/en/simatic-iot2020/pages/default.aspx
- https://www.udoo.org/

Intrebari

?

